Metal cutting and metrology

Q.1 What do you understand by machining of materials? Explain relevent theories?

Q.2 Draw in detail the Geometrical parameters of a single point cutting tool?

Q.3 Discuss theory of chip formation and show with diagram the various terms in chip formation.

Q.4 explain various type of chips formed during machining operation? Differential between continous and Discontinuous chip.

Q.5 What do you understand with cutting parameters. Explain briefly.

Q.6 Discuss the role of temp in machining.

Q.7 Discuss the various forces associated in orthogonal cutting with relevant diagram.

Q.8 When turning a 60 mm dia bar, it was observed that the tangential cutting forces are 3000 N and feed force is 1200 N. If the angle is 32°

Q. ① **Machining** :-

Machining is a manufacturing process in which excess material is removed from a workpiece in the form of chips using a cutting tool to obtain require shape, size and surface finish.

Ex → Turning, Milling, Drilling, shaping, Grinding.

Theories of machining

① **Plastic deformation theory** :-

- Material is removed by plastic flow along a shear plane.
- Mostly applicable to ductile materials (mild steel, aluminium).

② **Shear theory** :-

- Chip formation occurs due to shearing action along a shear plane.
- Cutting forces causes shear stress > shear strength of material.

③ **fracture theory** :-

- used for brittle materials.
- Materials breaks by cracking instead of plastic flow.

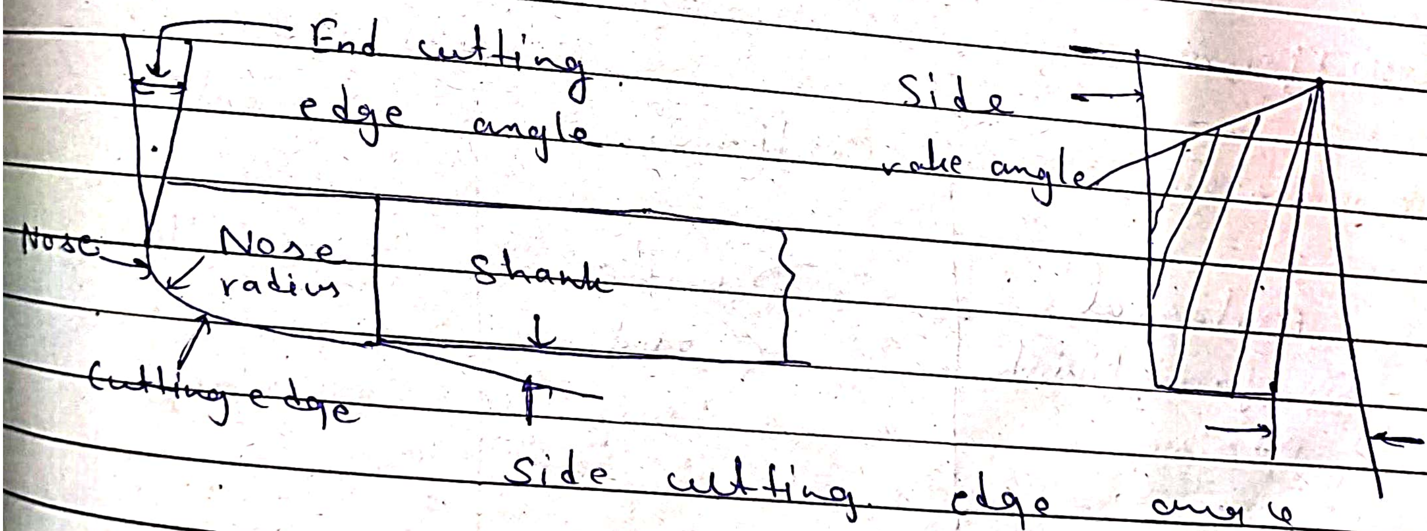
Q. 2

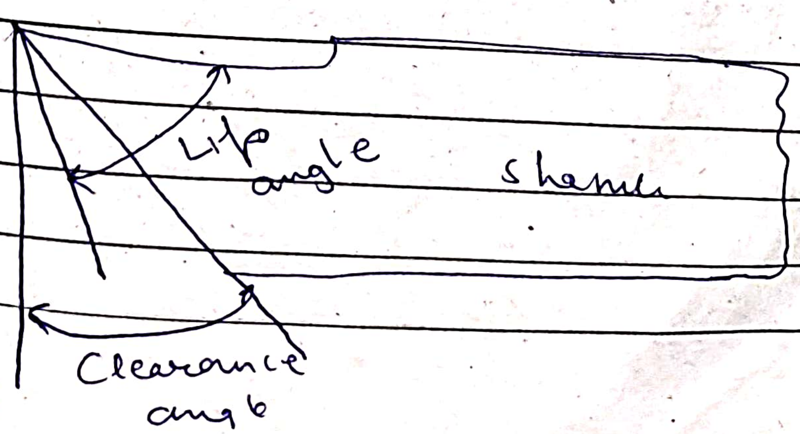
Single point cutting tool angles

1. Rake angle (α) – Helps in smooth chip flow.
2. Clearance angle (γ) – Prevents rubbing with work surface.
3. Cutting edge angle – Orientation of tool edge.
4. Nose radius – Improves surface finish.
5. Lip angle (β) – strength of cutting edge.

Functions

- Proper angle → less cutting force.
- Better surface finish
- Longer tool life.





Q. ④

Types of chips.

① Continuous chip.
- Ductile material.
- High cutting speed.
- Smooth surface finish.

② Discontinuous chip.
- Brittle materials.
- Low cutting speed.
- Rough surface.

③ Continuous chip with Built-up edge.
- Medium cutting speed.
- Poor surface finish.

Difference.

Feature	Continuous	Discontinuous
Material	Ductile.	Brittle
Surface finish.	Good.	Poor
Noise	Low	high.

Q. 5

Cutting parameters.

① Cutting speed (v)
Distance travelled by tool per min

$$v = \frac{\pi D N}{1000}$$

② Feed (f)
Tool movement per revolution (mm/rev)

③ Depth of cut (d)
Thickness of material removed (mm)

Effect

- Higher speed → more temp.
- Higher feed → rough finish
- Higher depth → higher cutting force.

Q. 6

Source of heat

- Plastic deformation in shear zone.
- Friction b/w tool and chip.
- Friction b/w tool and workpiece.

Effects

- Tool wear increases.
- Tool life decrease.
- Dimensional inaccuracy.
- Thermal damage to surface.

Control.

- Cutting fluids.
- Proper cutting parameters.
- Tool material selection.

Q. 7

Ans Forces in orthogonal cutting

1. cutting force (F_c) → Main force, along cutting dirⁿ.

2. Thrust force (F_t) → Perpendicular to cutting dirⁿ.

Resultant force (R)

$$R = \sqrt{F_c^2 + F_t^2}$$

Important

- Power calculation.
- Tool design.
- machine tool rigidity.

a.(v)

Theory of chip formation.

During a machining operation, when a single point cutting tool moves relative to the workpiece, the material ahead of the cutting ~~angle~~ edge is severly stressed.
When this stress exceeds the shear strength of the material, plastic deformation takes place and the material shears along a plane called shear plane, forming a chip.
This process is mainly explained by merchant's shear theory, which is applicable to ductile materials like mild steel and aluminium.

Stages of chip formation.

1. Compression of material ahead of tool.
2. Plastic deformation of material.
3. Shearing along shear plane.
4. Chip flows upward along rake face of tool.

Important terms used in chip formation.

1. Shear plane
   - Imaginary plane along which the material shears.
   - Makes angle ϕ (shear angle) with cutting dir^n.

2. Shear angle (α)
   - Angle b/w shear plane and cutting velocity diagram.
   - Higher shear angle $\rightarrow$ less cutting force

3. Rake angle (α)
   - Angle b/w rake face of tool and normal to work surface.
   - Helps in smooth chip flow.

4. Chip thickness
   - uncut chip thickness (t_1): Thickness of material before cutting
   - Cut chip thickness (t_2): thickness of chip after cutting.

$$t_2 > t_1$$

5. Cutting velocity (v)

· Velocity at which tool cuts the w/p.

6. Friction angle (β).

· Angle related to friction b/w tool and chip.

Merchant's shear angle relation.

$$\boxed{\phi = 45° + \frac{\alpha}{2} - \frac{\beta}{2}}$$

where;

ϕ = shear angle

α = rake angle

β = friction angle

Given;

Tangential cutting force $F_c = 3000$ N.

Feed (thrust) force $F_t = 1200$ N

Tool rake angle, $\alpha = 32°$

$\sin 32° = 0.530$
$\cos 32° = 0.848$

① Friction force on rake face.

$F = F_c \sin\alpha + F_t \cos\alpha$

$= (300 \times 0.530) + (1200 \times 0.848)$

$F = 2607.6$ N.

② Normal force. $N \Rightarrow N = F_c \cos\alpha - F_t \sin\alpha$.

$N = (3000 \times 0.848) - (1200 \times 0.530)$

$N = 1908$ N.

$\mu = \frac{F}{N} = \frac{2607.6 \text{ N}}{1908} \Rightarrow 1.37$

Q For a plate moving in a fluid b/w two parallel surfaces with different gap distances h_1, h_1

$$F = \mu A v \left(\frac{1}{h_1} + \frac{1}{h_2} \right)$$

$$= 0.60 \times 0.50 \times 0.6 \left(\frac{1}{0.008} + \frac{1}{0.012} \right)$$

$$= 37.5 \text{ N}$$

Given, $S_{yt} = 380 \text{ N/mm}^2$

$d = 50 \text{ mm}$

$t = 15 \text{ mm}$

$f(s) = 6$

Step I Perminible strem.

$$\sigma_c = \frac{S_{yt}}{(fs)} = \frac{380 \times 2}{6} = 126.67 \text{ N/mm}^2$$

$$\tau = \frac{S_{sy}}{(fs)} = \frac{0.5 \times S_{yt}}{(fs)} = \frac{0.5(380)}{6} = 31.67 \text{ N/mm}^2$$

$$\sigma_t = \frac{S_{yt}}{(fs)} = \frac{380}{6} = 63.33 \text{ N/mm}^2$$

Step II Load acting on rods.

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$$P = \frac{\pi}{4} d^2 \sigma_t$$

$$P = \frac{3.14}{4} (50)^2 \times 63.33$$

$$= 124388.16\ N.$$

Step III Inside diameter of socket =

$$P = \left[\frac{\pi}{4} d_2^2 - d_2 t\right] \sigma_t$$

$$124388.16 = \left[\frac{3.14}{4} d_2^2 - \frac{d_2 t}{1}\right] 63.33$$

$$1964 = \left[\frac{3.14 d_2^2 - 4 d_2 t}{4}\right]$$

$$7,856 = 3.14 d^2 - 4 d_2 \times 15.$$

$$= 3.14 d^2 - 60 d_2$$

$$d_2 = 65\ mm$$

Step IV Outside diameter of socket (d_1).

$$P = \left[\frac{\pi}{4}(D_1^2 - d_1^2) - (D_1 - d_1)t\right] \sigma_t$$

$$\frac{124388.16}{63.33} = \left[\frac{\pi}{4}(D_1^2 - d_1^2) - (D - d_1)t\right]$$

$$1964 = \frac{\pi}{4}(D_1^2 - 65^2) - (d_1 - 65)(15)\Big]$$

Q $P = 150 \times 10^3$ N
$fs = 5$

Calculation of permissible stress

$$\sigma_t = \frac{S_{yt}}{fs} = \frac{400}{5} = 80\ N/mm^2$$

$$\sigma_c = \frac{S_{yc}}{(fs)} = 80\ N/mm^2$$

$$\tau = \frac{S_{sy}}{(fs)} = \frac{0.5 S_{yt}}{fs} = 40\ N/mm^2$$

Calculation of dimensions

$$D = \sqrt{\frac{4P}{\pi \sigma_t}} = \sqrt{\frac{4 \times 50 \times 10^3}{3.14 \times 80}}$$

$$= \sqrt{\frac{200,000}{251.2}} = \sqrt{79.61}$$

$$D = 30\ mm$$

Enlarged dia of rods (D_1)

$$D_1 = 1.1D = 1.1(30) = 33 \text{ or } 35\ m$$

Dimensions of a and b.

$$a = 0.75D = 0.75(30) = 22.5 \text{ or } 25\ mm$$
$$b = 1.25D = 1.25(30) = 37.5 = 40\ m$$

Diameter of pin $= \sqrt{\frac{2P}{\pi \tau}}$

$$= \sqrt{\frac{2(50 \times 10^3)}{\pi (40)}} = 28.21 \text{ or } 30\,mm$$

stress in eey

$$\sigma_t = \frac{P}{b(d_o - d)} = \frac{50 \times 10^3}{40 \times (80 - 40)} = 31.25\,N/mm$$

Q

Unit → 2

Metal cutting and metrology

Q. 1 What is stick - slip phenomena ?

Ans Stick - Slip phenomena is the condition during metal cutting in which the ~~slip~~ chip alternately sticks to and slip over the tool face, causing fluctuation in cutting force and poor surface finish.

Q. 2 Define tool wear ?

Ans Tool wear is the gradual ~~tool~~ loss of material from the cutting edge of a tool during the machining process.

Q 3 what is tool life ?

Ans Tool life is the time period for which a cutting tool works satisfactorily before it requires re - sharpening or replacement.

Q. 4 write taylor's tool life eqn ?

$$\boxed{VT^n = C}$$

where,

V = cutting speed

T = tool life

n & c = constants

Factor's affecting tool life.

① Cutting speed – Inc. in cutting speed reduces tool life.

② Feed and depth of cut – Higher feed and depth inc. tool wear.

③ Tool material – Harder tool material give longer tool life.

④ Workpiece material – Harder material reduces tool life.

Q Explain crater wear and its effects?

Crater wear occurs on the rake face of the cutting tool due to continuous sliding of hot chips over the tool surface.

Causes :-

- High cutting temp.
- Diffusion of tool material into the chip.
- High cutting speed.

Effects :-

- Weakening of cutting edge.
- Inc in cutting forces.
- Reduction in tool life.

Q Explain flank wear with neat sketch ?

Flank wear is the wear that occurs on the flank face of the tool due to rubbing b/w the tool and the machined surface.

Causes :-

- Abrasion by hard particles.
- High cutting temp.
- Continuous contact with w/p.

Effects :-

- Poor surface finish.
- Dimensional inaccuracy.
- Increased power consubtion.

Q Explain adhesion and diffusion wear

Adhesion wear :-

Adhesion wear occurs. When tool and w/p materials weld together locally under high pressure and temp, and fregments are torn from the tool.

Effect :- Tool material loss and poor surface finish

Diffusion wear :-

Diffusion wear occurs when tool atoms diffuse into the chip or w/p at high temp.

Effect : Weakening of the tool at high cutting speeds.

Q Explain the effect of cutting speed on tool life.

As cutting speed inc., cutting temp inc which accelerates tool wear and reduces tool life.

$$V T^n = C$$

This shows that tool life decreases rapidly with inc. in cutting speed

Tool wear types
- Crater wear
- flank wear.

Tool wear mechanism

- Abrasion
- Addhesion
- Diffusion
- Oxidation.

Cutting Temp :-

Cutting temp is the temp genrated at the tool - chip - workpiece interface during the metal cutting process.

Effects → of cutting.

1. Reduction in tool life.
2. Poor surface finish.
3. Dimensional inaccuracy.
4. Tool material softening
5. Inc. in power consumption.